

Interpreting YOLO-based Object Detection in Autonomous Driving via XAI Methods

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Abstract

Object detection is a critical component of autonomous driving systems, where reliability and interpretability are essential under diverse environmental conditions. Despite recent advances in YOLO-based detectors achieving high accuracy in challenging scenarios such as nighttime and rain, their decision-making processes remain largely opaque. In this study, we investigate the explainability of a pretrained YOLOv8 object detector using gradient-based (Grad-CAM-style) and perturbation-based (RISE) methods. We analyze how visual evidence supporting detections changes across daytime, nighttime, and rainy scenes, and quantitatively evaluate explanation faithfulness using insertion and deletion games. Our results show that while salient evidence remains causally relevant across conditions, adverse visibility leads to more spatially distributed reliance on contextual cues. This work highlights the importance of combining qualitative and quantitative analysis to better understand object detector behavior in safety-critical settings.

1. Introduction

Object detection models are widely used in autonomous driving, where environmental conditions such as lighting and weather significantly affect visual perception. Although modern detectors achieve high accuracy, their decision-making processes remain difficult to interpret, particularly under adverse conditions such as nighttime or rain.

In this project, we analyze the explainability of a pretrained YOLOv8 object detector using gradient-based (Grad-CAM-style) and perturbation-based (RISE) methods. We examine how visual evidence supporting detections varies across daytime, nighttime, and rainy scenes, and quantitatively assess explanation faithfulness using insertion and deletion games to evaluate causal relevance.

2. Experimental Setup

2.1. Dataset and Detection Model

The evaluation dataset consists of street-scene images grouped into three visibility conditions: daytime, nighttime, and rainy environments, representing increasing levels of visual difficulty. The dataset is intentionally kept small to enable detailed, instance-level explainability analysis rather than large-scale performance benchmarking.

All images are resized to a fixed resolution of 640×384 pixels, with no additional preprocessing applied. A pretrained YOLOv8n model provided by the Ultralytics framework is used without fine-tuning to ensure that observed explanation patterns reflect inherent model behavior. Since YOLOv8 produces multiple detections per image, a single high-confidence vehicle detection is selected and fixed as the target across all explainability and faithfulness experiments.

2.2. Grad-CAM-style Explanation for Object Detection

Grad-CAM is a widely used gradient-based explanation method originally proposed for image classification models. It identifies important image regions by analyzing gradients flowing into convolutional feature maps and produces a spatial heatmap highlighting areas that most strongly influence a prediction.

Applying Grad-CAM directly to object detection models is not straightforward. Unlike classification models that output a single class score, YOLOv8 produces multiple bounding boxes across different spatial locations and detection heads. As a result, there is no single scalar output that directly corresponds to a specific detection of interest.

To address this challenge, a Grad-CAM-style approach tailored to object detection is adopted. Intermediate feature maps are captured by registering forward hooks on the final convolutional layer of the YOLOv8 network. Instead of using a class logit, a detection-specific score is defined by aggregating feature activations within the spatial region corresponding to the selected target bounding box. Gradients of

this score are then backpropagated to compute channel-wise importance weights.

These weights are used to generate a heatmap that reflects the internal sensitivity of the network to different image regions with respect to the target detection. While this approach differs from the original Grad-CAM formulation, it preserves the core principle of gradient-based localization and is commonly used in explainability studies for object detection models. The resulting heatmaps provide insight into how internal feature representations contribute to individual detections.

2.3. RISE: Perturbation-based Explanation

To complement gradient-based explanations, a perturbation-based explanation method, RISE, is employed. Unlike Grad-CAM, which relies on internal gradients, RISE estimates pixel importance by observing how random input perturbations affect the model’s output. This makes RISE model-agnostic and suitable for analyzing complex architectures such as object detectors.

RISE generates a large number of random binary masks that occlude different regions of the input image. Each masked image is passed through the detector, and the contribution of each mask is weighted by the resulting detection confidence. The final explanation heatmap is obtained by aggregating these weighted masks, producing an estimate of behavioral importance.

Applying RISE to object detection introduces additional challenges. Masking parts of the image can cause the detector to shift focus to different objects or produce new detections altogether. To mitigate this instability, the target detection is fixed using Intersection-over-Union (IoU) matching. At each perturbation step, only detections that sufficiently overlap with the original target bounding box are considered, and the confidence of the best-matching detection is tracked.

By fixing the target detection in this manner, RISE explanations can be interpreted consistently across perturbations. This approach allows RISE to capture how different regions of the image causally influence the detector’s confidence for a specific object, providing a complementary perspective to gradient-based explanations. This design ensures that changes in detection confidence are attributable to input perturbations rather than shifts in detection targets.

3. Results

Figure 1 presents qualitative comparisons of detection results and explanation maps across different visibility conditions.

3.1. Qualitative Results

Qualitative comparisons across visibility conditions reveal clear differences in explanation behavior. In daytime

scenes, Grad-CAM-style explanations tend to be relatively localized around the target object, such as a vehicle, indicating that under favorable conditions the detector relies primarily on object-centric visual cues.

In nighttime scenes, Grad-CAM heatmaps become more spatially diffuse. Attention often extends to surrounding regions influenced by artificial lighting, headlights, and shadows. This suggests that under low-light conditions, the detector relies on a broader set of visual cues beyond the object’s immediate contours.

Rainy scenes exhibit the most diffuse explanations. Reduced contrast, reflections on wet surfaces, and visual noise cause both Grad-CAM and RISE explanations to spread across larger portions of the image. In these cases, the detector appears to rely heavily on contextual information rather than precise object boundaries. RISE explanations are consistently more diffuse than Grad-CAM across all conditions, reflecting the perturbation-based nature of the method and its sensitivity to global context.

3.2. Insertion and Deletion Games

To quantitatively evaluate explanation faithfulness, insertion and deletion games are conducted using RISE importance maps. These games assess whether pixels identified as salient by the explanation method causally influence the detector’s decision for a fixed target object.

In the deletion game, pixels are progressively removed from the original image in descending order of importance. At each step, YOLO inference is performed and the confidence of the IoU-matched target detection is recorded. A rapid decrease in confidence indicates that the removed pixels contain critical evidence supporting the detection.

In the insertion game, the process is reversed. Starting from a blank image, pixels are gradually added back according to the same importance ranking. As informative regions are revealed, the confidence of the target detection is expected to increase.

For representative daytime and nighttime samples, the target detection confidence drops sharply under deletion and recovers rapidly under insertion. After normalization by the initial confidence, the daytime sample (day_05) exhibits a deletion AUC of 0.006 and an insertion AUC of 0.987, with confidence falling below 10% within two deletion steps and exceeding 90% within three insertion steps. In contrast, the nighttime sample (night_03) shows a higher deletion AUC of 0.011 and a lower insertion AUC of 0.962, requiring more steps for both degradation and recovery. These results indicate that, while salient evidence identified by RISE remains causally relevant in both conditions, evidence in nighttime scenes is less immediately removable and recoverable, consistent with a more distributed reliance on visual cues.



Figure 1. Qualitative comparison of YOLOv8 detections and Grad-CAM-style explanations under different visibility conditions. Each panel shows (from left to right) the original image, YOLOv8 detection output, and the Grad-CAM-style heatmap for a fixed target detection. The explanations become more spatially diffuse in nighttime and rainy scenes, suggesting increased reliance on contextual cues under degraded visibility.



Figure 2. RISE explanations with a fixed target bounding box. The heatmap is obtained by aggregating random binary masks weighted by the confidence of the IoU-matched target detection, and the selected box (white) is overlaid for reference. Across conditions, the spatial concentration of RISE explanations varies: in the nighttime example, the highest relevance is concentrated on the target vehicle, whereas the daytime and rainy examples exhibit more spatially distributed relevance, reflecting a stronger influence of contextual cues.

4. Discussion

The results highlight several important characteristics of explainability in object detection. First, environmental

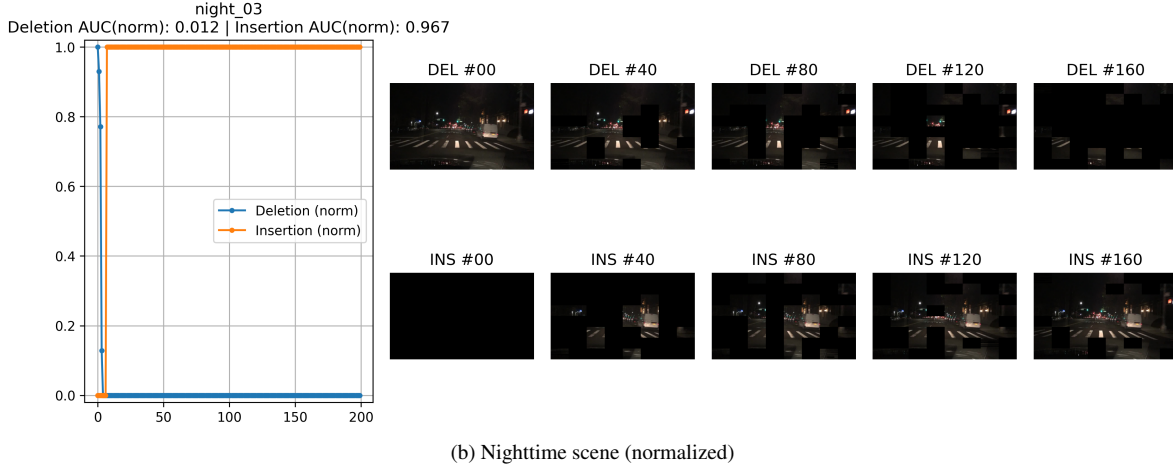
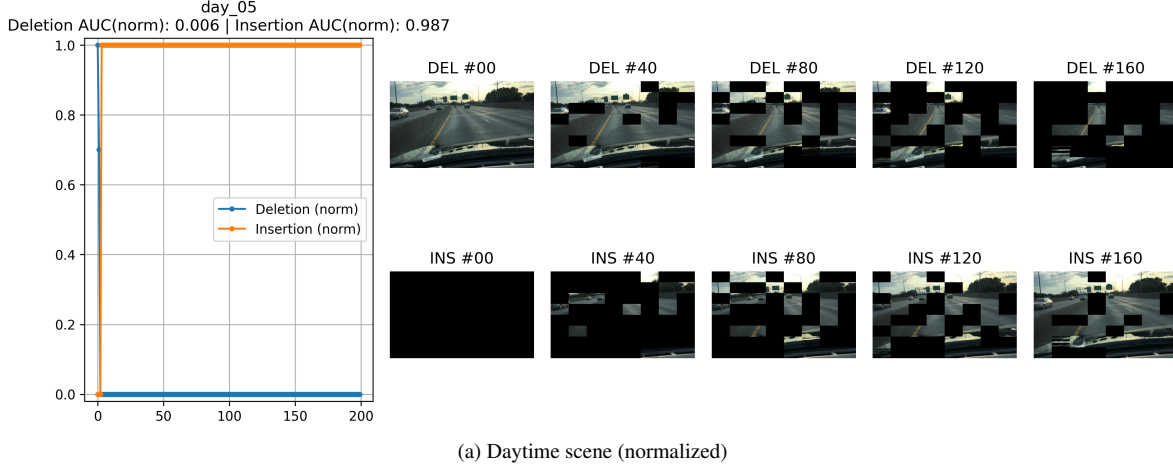


Figure 3. Normalized insertion and deletion game results for RISE explanations. For each condition, curves show the relative change in IoU-matched target detection confidence as salient pixels are progressively removed (deletion) or added (insertion). Normalization allows direct comparison of curve shapes as a faithfulness sanity check, independent of absolute confidence values.

conditions strongly influence the spatial structure of explanations. As visibility degrades, explanations become increasingly diffuse, suggesting a shift from object-centric to context-dependent decision-making.

Second, gradient-based and perturbation-based explanation methods capture different aspects of model behavior. Grad-CAM-style explanations reveal internal sensitivity patterns within the network, while RISE captures behavioral importance through input perturbations. The discrepancy between the two methods is informative rather than contradictory, emphasizing the need for multiple perspectives in explainability analysis.

Finally, the insertion and deletion games provide quantitative support for the qualitative observations. Rapid confidence degradation during deletion and recovery during insertion indicate that the explanations capture evidence that meaningfully affects the detector’s decisions, even though

this evidence becomes more distributed under adverse conditions.

5. Conclusion

This project analyzed the explainability of a YOLOv8 object detector under varying visibility conditions using both gradient-based and perturbation-based methods. The results demonstrate that environmental conditions significantly affect how detection models utilize visual evidence, with increasing reliance on contextual cues under nighttime and rainy scenarios.

By combining qualitative visualization with quantitative faithfulness evaluation, this study provides a more comprehensive understanding of object detector behavior. Future work could extend this to larger datasets, additional XAI methods, or alternative target selection strategies, further improving the interpretability of object detection systems.

5.1. References

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References

- [1] Alain Andres, Aitor Martinez-Seras, Ibai Laña, and Javier Del Ser. On the black-box explainability of object detection models for safe and trustworthy industrial applications. *Results in Engineering*, 24:103498, Dec. 2024. 5
- [2] Santiago Calderón-Peña, Hana Chockler, and David A. Kelly. Real-time incremental explanations for object detectors in autonomous driving, 2025. 5
- [3] Carman Gheorghe. Analyzing real-time object detection with yolo algorithm in automotive applications: A review. Tech-SciencePress, 2024. Accessed: 2025-10-25. 5
- [4] Loc Phuc Truong Nguyen, Hung Truong Thanh Nguyen, and Hung Cao. Odexai: A comprehensive object detection explainable ai evaluation, 2025. 5
- [5] Yuka Ogino, Yuho Shoji, Takahiro Toizumi, and Atsushi Ito. Erup-yolo: Enhancing object detection robustness for adverse weather condition by unified image-adaptive processing, 2024. 5
- [6] Shahriar Soudeep, Md Abrar Jahin, and M. F. Mridha. Interpretable dynamic graph neural networks for small occluded object detection and tracking, 2025. 5
- [7] Stavros Theocharis. easy-explain: Explainable ai for yolov8. Medium, 2024. Accessed: 2025-10-25. 5